

ECONOMIC APPLICATIONS OF NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

GUIDE TO THE ECONOMIC APPLICATIONS

This document is the current economic-applications companion to *Neural Bellman Operators*. It retains the complete application text of the reviewed article and supplement, including model primitives, boundary and utility restrictions, analytical derivations, equilibrium conditions, implemented numerical procedures, and their recorded successes and failures. It is organized by economic problem. Its section, equation, table, and theorem numbers are distinct from those in the main article and technical supplement; cross-references connect all three documents.

The applications share a policy-evaluation and feasible-improvement construction. For a given policy, the critic solves that policy's continuation equation under the stated terminal, exit, or reflection conditions. The actor changes only the actions whose deviations define the relevant economic problem. A control problem compares feasible policies, a sophisticated consumer compares local deviations against future selves, and a firm compares complete unilateral deviations against fixed rivals. A small fitted residual is not substituted for any of these economic criteria.

The homothetic portfolio and recursive tests use a known invariant representation and report its scope explicitly. The endogenous-preference application retains the role of cardinal normalization across controlled preference states, the reflected preference domain, the cross derivative in portfolio choice, and the envelope comparison for adjustment costs. Its continuous-action records retain actor failures, local-search contributions, whole-domain off-grid failures, fresh-grid training, and the separate consumption-compensation experiment. The temporal-self condition concerns infinitesimal deviations; the game calculations retain missing pure fitted stage equilibria and approximate best-response bounds. The coupled linear-quadratic laboratory has an exactly representable structure. These distinctions are substantive economic restrictions rather than interchangeable numerical labels.

TABLE A.1
ECONOMIC OBJECTS IN THE APPLICATION COMPANION

Problem	Continuation object	Economic assessment
Portfolio choice	Time-additive policy value	Analytical homothetic target and constrained actions
Recursive utility	The policy's own utility process	Utility domain, matching bequest, and comparison
Preference adjustment	Joint internal and wealth-state value	Consumption, portfolio, and adjustment choices; complete feasible-action accounts
Temporal selves	Two continuation components under a common future policy	Local spike-deviation equilibrium
Strategic investment	A distinct continuation value for each firm	Complete unilateral best response against fixed rivals
Linear-quadratic and capital laboratories	Coupled policy values and costates	Explicit representation tests, independent payoff and residual diagnostics

General approximation results and proofs are in the current technical supplement, beginning with Section S.5. The nonlinear capital economy and its current continuation-learning comparison are in the main article. The material below reproduces the complete application source with relocated input paths and documented grammar corrections; its reported quantities retain their original model, sample, and numerical meaning.

A.1. COMPLETE ECONOMIC APPLICATIONS

A.1.1. *Analytical and Nonsmooth Benchmarks*

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (\text{A.1})$$

With $\rho = 0.04$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$, these numbers are $p^* = 0.75$ and $m^* = 0.04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule. A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = 0.01$ with no short sales, the unconstrained share is -0.125 and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = 0.03$. The maximized HJB remains an equality. The action KKT condition is an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section S.5, rather than this homothetic corner solution, supplies the nonsmooth selection test.

A.1.1.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates 0.025 and 0.008; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0, 1, 2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that finite-horizon

TABLE A.2

CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	0.04125	0.75000	3	5.6×10^{-16}
No-short-sale corner	0.03000	0.00000	3	2.3×10^{-16}
Recursive matching-bequest test	0.04550	0.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

interpretation, not a boundary condition hard-imposed during these particular updates.

Table A.2 reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within 3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

A.1.2. Recursive Utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (\text{A.2})$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (\text{A.3})$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This

distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (\text{A.4})$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2 \gamma \sigma^2} \right], \quad K = [\rho (m^*)^{a-1}]^{b/a}. \quad (\text{A.5})$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = 0.04$, $\mu = 0.08$, $r = 0.02$, $\sigma = 0.2$, this gives $(m^*, p^*) = (0.0455, 0.3)$.

A.1.2.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (A.5) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[\underline{x}, \bar{x}]$ with $0 < \underline{x} < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping boundary and at T . Consumption shares are in a compact positive interval containing 0.0455, and portfolio shares are in a compact interval containing 0.3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the Hamiltonian and matches all stopping payoffs; Theorem S.1 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-complementarity parameters straddle one ([Herdegen et al., 2021](#)). The paper retains

that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

A.1.2.2. *Stability and baseline design*

The recursive experiment in Table A.2 tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript's constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this paper. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

A.1.3. *Endogenous Preferences and Portfolio Choice*

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \quad (\text{A.6})$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (\text{A.7})$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses $[1.2, 3]$, so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative or transformed diffusion, but that would be a different primitive and must be identified as such.

A.1.3.1. Cardinal utility and the objective

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (\text{A.8})$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (\text{A.9})$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (\text{A.10})$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately 0.84657. This is a derivative of flow utility, not a sign theorem for the endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

A.1.3.2. Hamiltonian, constrained policies, and economic implications

The HJB equation is

$$\begin{aligned}
 -V_t = \sup_{c,p,\theta} \{ & U(c,u) - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c]V_X \\
 & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}p^2\sigma_X^2 X^2 V_{XX} + \varrho\sigma_u\sigma_X p X V_{uX} \}. \quad (\text{A.11})
 \end{aligned}$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (\text{A.12})$$

$$\theta^* = V_u/k, \quad (\text{A.13})$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (\text{A.14})$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (A.13) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2}\mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (\text{A.15})$$

The sign of the hedging component in (A.14) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

A.1.3.3. *Independent two-state reference and identification*

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[0.5, 2]$, and $G(u, X) = X^{1-u} / (1 - u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \varrho, \rho) = (0.02, 0.08, 0.2, 0.08, -0.3, 0.04)$. The finite action set is the product of $c/X \in \{0.02, 0.05, 0.1\}$, $p \in \{0, 0.375, 0.75\}$, and $\theta \in \{-0.15, 0, 0.15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (0.1, 0.75, -0.15)$ at that point. The approximately 0.048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section A.1.3.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \theta) + \varepsilon_j$, let $m(\theta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_\theta m$, after accounting for latent preference states, normalization parameters, and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

The next comparison is the retained continuous-action study evidence under its original enclosure. The new signed envelope and fresh nested-grid policies are reported separately in Section A.1.8; historical failure counts are not overwritten.

A.1.3.4. *Neural preference policies and the role of the action oracle*

We compute the original preference-adjustment economy with the cardinal utility, shock correlation, reflected preference state, and wealth stopping payoff specified above. The main grid has 17 preference points, 25 log-wealth points, and 20 time steps over a unit horizon. Adjustment cost is $k = 1$. The expanded control box is

$$m = c/X \in [0.02, 0.30], \quad p \in [0, 1.5], \quad \theta \in [-0.45, 0.45].$$

The finite-action comparison uses the 175 triples formed by five consumption shares, five portfolio shares, and seven adjustment efforts, including all endpoints. This is the same constrained economy across the matched finite methods, not a change of utility normalization.

Two experiments distinguish action representation from action optimization. First, the finite-policy study trains two-hidden-layer, width-32 tanh critics and categorical actors. At each backward step it computes the complete 175-action payoff vector using the current continuation critic. Its exact argmax provides a label, and the vector also supplies the actor’s expected action-loss objective. Thus *the training oracle is exhaustive*, even when the final actor is accurate. It is a useful neural policy-representation experiment, but not evidence that the actor avoided Bellman maximization.

Second, the continuous-action study trains a continuous-output actor directly through its own positive-weight payoff. It never receives an all-action value vector or an exact argmax label. The actor receives 160 Adam updates per time level; the continuation critic receives 320 Adam updates and up to 300 L-BFGS iterations, with failed polishing steps rolled back. The actor-free direct comparator optimizes eight independent bounded controls at every state for the same 160 iterations, followed by the same critic fit. Every restart and rejected action query is charged. Neither procedure invokes a Riccati solution, an analytically invariant policy class, or the independently computed reference during training.

Finite-action neural policies. The raw finite-policy study actor is the primary neural representation result. Its full dynamic policy loss is below the declared 0.1 target for each fixed seed. A top-four shortlist and a separately reported correction

map further reduce the loss. The correction map is not part of the raw network. The complete finite policy, not only one-step deviations against a fitted critic, is evaluated by independent backward recursion.

TABLE A.3

FINITE PREFERENCE POLICIES: FINITE-POLICY STUDY EVIDENCE

Method	Seed	Raw loss	Final loss	Value error	Seconds	Guard (%)
NBO	11	0.0445	0.0261	0.0924	30.59	0.294
Direct	11	0.0181	0.0181	0.0790	24.79	0.000
NBO	29	0.0500	0.0260	0.0814	30.49	0.332
Direct	29	0.0263	0.0263	0.0551	25.19	0.000
NBO	47	0.0500	0.0263	0.1325	29.89	0.230
Direct	47	0.0264	0.0264	0.0824	25.17	0.000

Loss bounds cover every stored state and time of the same 175-action economy. Seconds include training and the three reported policy verifications; classical backward induction averages 0.102 seconds. A raw actor, shortlist, and corrected policy are distinct objects. Utility losses are absolute units.

These results repair the earlier initial evaluation study actor failures, which remain in the historical numerical record. They do not reverse the cost comparison: the matched direct finite-Bellman fit is faster and more accurate on average, and classical backward induction is substantially cheaper on this small grid. Tables report those comparisons without selecting a favorable seed. The actor’s approximation cost is part of the result, not an argument for discarding the economic application.

Continuous-action optimization without exact labels. The initial continuous actor and direct-gradient pilot fail the 0.1 deviation diagnostic. The actor can saturate at a portfolio bound while a lower portfolio share improves continuation value. Four actor heads with output reinitialization do not cure that failure in the retained pilot. We therefore execute a disclosed local-improvement variant: compare the actor proposal with eight box corners, then perform two coordinate-search sweeps at each of eight decreasing step sizes. The same local search is applied to the direct

comparator. An actor-free ablation omits all gradient action updates and starts the identical search from feasible proposals. This ablation measures how much of the hybrid result comes from the search itself.

TABLE A.4

CONTINUOUS-CONTROL TRAINING WITHOUT EXACT MAXIMIZING LABELS

Method	Seed	Raw diagnostic	Hybrid gain	Train (s)	Queries (10^6)
Actor + search	11	0.2596	0.0617	15.01	2.261
Actor + search	29	0.0733	0.0188	28.49	2.261
Actor + search	47	0.2597	0.0475	26.46	2.261
Direct + search	11	0.1554	0.0303	32.02	11.841
Direct + search	29	0.1566	0.0203	32.80	11.841
Direct + search	47	0.1554	0.0251	30.52	11.841
Search only	11	5.5370	0.0617	13.97	0.961
Search only	29	7.2093	0.0616	13.72	0.961
Search only	47	8.0316	0.0616	12.87	0.961

Raw diagnostic is the largest finite-reference payoff minus the raw policy payoff. Hybrid gain is the augmented finite-deviation gain, a lower bound on continuous-action regret, not a certificate. All methods use zero exhaustive training backups and zero exact argmax training labels. Search-only omits action-gradient updates. Training, reference, and full-cover verification costs are separate.

The table distinguishes three quantities. The raw-network loss against the finite reference tests the actor alone. The augmented finite-deviation gain permits all inherited actions and the learned continuous action at every state and time; it is a *lower bound* on missed continuous-action gains. The complete action-box upper bound in Proposition S.7 is a separate verification result. A small augmented gain is not inserted into that proposition as if it were a uniform upper bound. Local changes occur at nearly every active state-time node in the principal hybrid runs, so this experiment does not describe a sparse correction map or an actor-only controller.

The action cover includes binding constraints and wealth exits. Its unresolved boxes remain in the bound when the declared budget is exhausted. The resulting upper array can audit the actor, the direct optimizer, and the local-search ablation without retraining. This produces an operational error account on a multidimen-

TABLE A.5

FULL CONTINUOUS-ACTION POLICY BOUNDS ON THE TWO-STATE NODAL ECONOMY

Policy	Seed	Uniform bound	Center bound	Eval. (s)	Bound ≤ 0.1
Actor + search	11	0.1294	0.0601	0.22	No
Actor + search	29	0.1227	0.0583	0.22	No
Actor + search	47	0.1301	0.0608	0.25	No
Direct + search	11	0.1153	0.0604	0.23	No
Direct + search	29	0.1156	0.0586	0.22	No
Direct + search	47	0.1203	0.0609	0.25	No
Search only	11	0.1196	0.0647	0.24	No
Search only	29	0.1179	0.0586	0.23	No
Search only	47	0.1209	0.0610	0.22	No

The full action box is covered at all 425 states and 20 times. The common optimal upper envelope costs 1624.59 seconds across the retained implementations and 40.96 million action-box evaluations. Each row adds its own interval policy evaluation. Unresolved leaves retain their upper bounds. No continuous-state/time error is set to zero.

sional state grid with a continuous, nonconcave action objective. It does not establish that all runs meet the tighter 0.1 complete-action target, nor that the actor dominates the direct method at matched economic accuracy.

Alternative representations and economic units. Additional same-model comparisons use a gated neural continuation architecture and tensor Chebyshev projection. The gated method uses DGM-style state-dependent gates (Sirignano and Spiliopoulos, 2018), but here it is fitted to finite Bellman targets, not called a mesh-free differential DGM solver. The projection exploits the same two-state geometry and imposes the same wealth-boundary lifting. Degrees, parameter counts, all-action costs, failures, and actual accuracy appear in the supplement. These are executed post-review comparisons, not results attributed to the earlier unexecuted development protocol. A generic semilinear BSDE comparator is not substituted for this controlled-covariance, stopped problem without reformulating its second-order control term.

All losses are in the specified cardinal utility units. The optimal finite-model value at the center is approximately -4.8410 , and the initial value range is approximately $[-9.1599, -0.1250]$. A loss of 0.1 is about 2.066 percent of the center's *absolute utility magnitude*; this scaling is not a consumption-equivalent welfare percentage. Endogenous curvature and adjustment costs make a common consumption-equivalent conversion a separate economic calculation.

Action bounds and controlled refinements. The sensitivity study retains both adjustment-cost values, the original 27-action truncation, 175 and 1,053 actions in the expanded box, a wider action box, wider wealth bounds, and finer space-time grids. Consumption and portfolio upper bounds remain frequently active. Expanding the box changes the constrained economy; refining actions inside one box does not. The continuous-action study additionally evaluates the *same frozen continuous policy*, bilinearly interpolated in state and held on its original time schedule, on three nested state-time levels. This isolates changes in policy evaluation from retraining. The supplemental differences are observed refinement diagnostics, not an estimated order promoted to a continuous-time error bound. The original continuous control problem, its utility normalization, and its comparative-static questions remain the economic objects of interest.

A.1.4. *Sophisticated Temporal Selves*

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (\text{A.16})$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (\text{A.17})$$

Write $V = \beta W_1 + (1 - \beta)W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (\text{A.18})$$

The current self's spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (\text{A.19})$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The appendix derives (A.19) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (A.18) and one actor for (A.19). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

A.1.4.1. An equilibrium calculation with endogenous continuation values

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$. Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1 - \beta}{\lambda_2(m)} \right]. \quad (\text{A.20})$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$. Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = 0.02$, $\rho = 0.04$, $\gamma = 2$, and $\kappa = 0.2$, the roots for $\beta = 1, 0.9, 0.7, 0.5$ are respectively 0.030000, 0.031372, 0.034606, and 0.038760. The code independently optimizes finite-duration deviations and records the gain divided by duration

as the duration shrinks. For $\beta = 0.7$, this quantity declines from approximately 4.60×10^{-5} at duration 0.1 to 4.66×10^{-9} at duration 0.001. This is a diagnostic of the derived spike equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

A.1.5. *Dynamic Games*

The dynamic Cournot application retains firm-specific capital and investment. Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in \mathcal{A}_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (\text{A.21})$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

A.1.5.1. *Player-specific differentiation and deviation bounds*

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_\nu \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2V_i)]. \quad (\text{A.22})$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem S.1 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem S.2 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is $1/3$ per firm. Joint-profit maximization gives a symmetric allocation $1/4$. At the latter profile, a firm optimally deviates to $3/8$, earning a gain of $1/64$. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

A.1.5.2. *Symmetry and market scaling*

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price

tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section A.1.5.3 replaces its evidentiary role with a fully specified finite dynamic game and independent best-response calculations, without asserting convergence of unrestricted neural game training.

A.1.5.3. *Neural dynamic capital policies and full unilateral deviations*

For two firms, we retain the capital game with $A = 1$, $P_0 = 2$, $\delta = 0.1$, $\sigma_K = 0.15$, $\rho = 0.04$, $T = 3$, investment in $[0, 0.8]$, and capital in $[0.05, 1.5]^2$. Terminal or joint-domain exit pays $0.5K_i$. Baseline adjustment coefficients are $b_1 = b_2 = 1$; the earlier asymmetric experiment has $b_2 = 1.2$. The principal neural grid has 17×17 states, 30 time steps, and 11 actions per player. Positive cubature and interpolation define the finite game. Market normalizations $M_2 = 1$ and $M_2 = 2$ are distinct economies, not evidence for an uncomputed large-population limit.

At each time and state, the finite-policy study calculation trains separate critics and actors. It constructs both full 11×11 continuation-payoff matrices. The training labels are a pure one-step equilibrium when one exists and a minimum-maximum-defect pair otherwise. The actor's own objective is differentiated with the rival action and continuation critic fixed. Thus this study actually trains neural strategic policies, but it also pays for an exhaustive joint-action training oracle. Raw policies, three-by-three shortlist policies, and any corrected policies are distinct stored arrays.

After freezing the entire profile, an independent backward optimization permits every own action at every state and time against the fixed rival policy. Its value minus the profile payoff is the unilateral exploitability map. This is not a one-period test with the approximate continuation held fixed. The raw and final profiles meet the 0.1 absolute-profit target in every principal finite-policy study case; the full-oracle guard is unused in those cases.

Five state-time nodes across the six neural runs lack a pure one-step equilibrium for the fitted continuation matrices. Their locations and defects are reported in the supplement. A minimum-defect selection at those nodes is not called an exact equilibrium. Nevertheless, the complete dynamic best-response bound remains

TABLE A.6

FINITE-POLICY STUDY NEURAL DYNAMIC-GAME PROFILES AND FULL BEST RESPONSES

Market	Seed	Raw max	Firm 1	Firm 2	Missing	Seconds
1	11	0.0259	0.0164	0.0252	3	52.43
2	11	0.0241	0.0255	0.0163	1	56.13
1	29	0.0300	0.0288	0.0296	0	52.44
2	29	0.0626	0.0624	0.0547	1	57.23
1	47	0.0190	0.0081	0.0183	0	52.53
2	47	0.0101	0.0074	0.0113	0	56.98

All bounds cover every stored finite state, time, and unilateral action, including deviations at the reported missing pure stages. The full-oracle guard is unused in these principal runs. Profit losses are absolute units; market normalizations are different economies.

valid for the resulting profile: its proof does not assume that every fitted one-step game has a pure solution. Moreover, for fixed Markov rivals, randomized or history-dependent own deviations cannot improve on the finite Markov best-response value, because each conditional action mixture is bounded by its largest pure-action payoff. The calculation therefore checks such unilateral deviations as well; it does not compute a mixed-strategy equilibrium profile or a continuous-action game.

The continuous-action study adds actor-free neural continuation fitting and classical backward selection for both market normalizations. The neural comparator allocates 480 critic Adam updates per time level rather than finite-policy study's 320 critic plus 160 actor updates, with the same L-BFGS cap. Counts and timings reveal the difference between nominal update matching and matched wall time. Both methods receive their own independent full dynamic best responses. The earlier six classical game cases, asymmetric costs, alternative tie rules, boundary frequencies, coarse counterexample, and finer-grid results remain in the supplement and raw record.

The finite-game result is a numerically evaluated approximate equilibrium with a specified profit-unit bound. It is not inferred from stationarity of a summed actor loss or the static 1/64 diagnostic. State interpolation, continuous investment,

TABLE A.7

CONTINUOUS-ACTION STUDY: ACTOR-FREE DYNAMIC-GAME COMPARISONS

Method	Market	Seed	Firm 1	Firm 2	Train (s)	Verify (s)
Classical	1	11	0.0000	0.0000	0.22	6.40
Classical	2	11	0.0000	0.0000	0.23	6.29
Direct	1	11	0.0223	0.0273	37.43	6.09
Direct	1	29	0.0199	0.0282	37.68	6.18
Direct	1	47	0.0223	0.0409	36.03	6.26
Direct	2	11	0.0091	0.0694	40.43	5.75
Direct	2	29	0.0306	0.0185	43.07	6.34
Direct	2	47	0.0206	0.0358	40.47	6.08

The direct method fits two continuation networks with the same class and 480 critic Adam updates, versus finite-policy study's 320 critic and 160 actor updates. Classical selection uses no neural approximation. Every method receives a new full dynamic best response. The finite-policy and continuous-action study timings were measured in their recorded environments; cross-environment ratios are not a hardware-controlled speed experiment.

diffusion timing, and continuously monitored exits remain separate terms in a continuous-game error account.

A.1.6. Scalability and the Diffusion Trace

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional components, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (\text{A.23})$$

The drift is a nonnormal dense matrix $A = -0.7I + 0.06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + 0.311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction

of the original benchmark is unavailable. All methods still exploit the declared quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P) / \rho. \quad (\text{A.24})$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

A.1.6.1. *Trained, coupled nonquadratic capital economies*

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [0.02, 2]$, and

$$\begin{aligned} dY_i = & \left[0.10 + 0.15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(0.15^2 + 0.10^2) \right] dt \\ & + 0.15 dW_i + 0.10 dW_0, \end{aligned} \quad (\text{A.25})$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - 0.10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed.

Horizon is one, discount is 0.04, and terminal utility is $\bar{Y} - 0.03d^{-1} \sum_i (Y_i - \bar{Y})^2$.

The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense nonlinear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + 0.2\bar{m}^*)^{-1}, 0.02, 2 \right),$$

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20$, 600 Adam updates, and batches of 128 independently sampled state-time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-0.5, 0.5]^d$. A common diagnostic target is residual root mean square at most 0.025 and maximum queried action gap at most 0.01. The recorded first crossing is not necessarily sustained; final failures are also reported.

The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verification box to $[-1.5, 0.5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

TABLE A.8

ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-0.5, 0.5]^d$. The failure criterion is RMS residual above 0.025 or queried action gap above 0.01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

A.1.6.2. Method-paired payoff evidence and arithmetic replay

The finite-policy study follow-up evaluates the coupled-capital policies at dimensions ten and twenty, for all three fixed seeds, both fitting domains, and 40, 80, and 160 Euler steps. The same 512 Brownian paths are used by NBO and the direct comparator within every cell. For path j , define $D_j = J_j^{\text{NBO}} - J_j^{\text{direct}}$. We report $\bar{D}$ and the pointwise Student interval $\bar{D} \pm t_{511, 0.975} s_D / \sqrt{512}$. Pairing removes the irrelevant component of common path variation from the comparison. The complete set of 36 comparisons is reproduced from the pinned raw arrays in the supplement.

The old twenty-dimensional fitting box is exited by essentially all recorded paths. Wider-domain fitting eliminates observed exits in the recorded audits and changes both payoff agreement and method differences. Zero observed exits in 512 paths is not a zero exit-probability theorem. Likewise, these intervals quantify pointwise Monte Carlo uncertainty under a specified Euler scheme. They are neither simultaneous confidence statements across all 36 comparisons nor bounds on time bias, optimal value, or the global differential residual. The experiment identifies a

TABLE A.9

 PAIRED PAYOFF DIFFERENCES AT THE FINEST RECORDED EULER LEVEL

Dim.	Seed	Domain	Difference	95% lower	95% upper
10	11	Narrow	-0.03338	-0.03544	-0.03133
10	11	Wide	0.00052	0.00049	0.00054
10	29	Narrow	-0.03094	-0.03250	-0.02938
10	29	Wide	-0.00003	-0.00004	-0.00002
10	47	Narrow	-0.02755	-0.02941	-0.02569
10	47	Wide	0.00000	-0.00001	0.00001
20	11	Narrow	-0.06302	-0.06572	-0.06033
20	11	Wide	0.00084	0.00080	0.00088
20	29	Narrow	-0.01449	-0.01490	-0.01408
20	29	Wide	0.00409	0.00394	0.00424
20	47	Narrow	-0.20161	-0.20211	-0.20111
20	47	Wide	0.00058	0.00053	0.00062

NBO minus direct payoff on the same 512 paths. Full 40/80/160-step comparisons are in the supplement. Confidence intervals quantify Monte Carlo variation only.

consequential coverage failure and its measured correction without redefining the unbounded economic state space as a stopped box.

The one-state consumption certificate has a different evidentiary status. finite-policy study independently replays its previously generated complete leaf cover with `mpmath.iv` at 40 decimal digits, without importing the original interval arithmetic implementation. For the stored seed-11 networks, the NBO policy-loss bound is about 0.04961 and the direct-HJB bound about 0.04967. Both meet the stated threshold. This is an *independent interval-arithmetic replay on the previously generated complete cover*, not independent network training, independent cover generation, or a formal machine proof. Together with the new complete three-control cover on the two-state preference grid, it demonstrates two different ways of operationalizing verification. Neither removes the dimensional cost of covering an unrestricted high-dimensional continuous state space.

A.1.6.3. *Exact and stochastic trace computation*

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (\text{A.26})$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (\text{A.27})$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (\text{A.28})$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics.

All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator's execution without identifying a random training objective with a deterministic certificate.

A.1.7. Validated Evaluation and Action Improvement

The neural Bellman map must account for two distinct approximation decisions: which continuation values are passed to the next stage, and how tightly the admissible action supremum is bounded. This section supplies an explicit output rule for the first decision and a signed enclosure for the second. Neither construction presumes that a stationary point of the training loss is economically optimal. Both apply to the original preference economy and leave its controls, state law, and stopping convention unchanged.

A.1.7.1. An evaluation rule with a prescribed nodal error

Let T_h^a denote a positive-weight, one-step policy operator on a finite state grid, and suppose its continuation weights have sum at most $q_h \leq 1$. At a given date, let w be the continuation vector, a the selected feasible policy, $\hat{v}$ the fitted neural critic, and $\tilde{T}_h^a w$ the computed selected-policy target. We distinguish the computation of this target from maximization over actions. In particular, it uses one action at each node, not a table of optimizing labels. For a prescribed tolerance $\tau_h > 0$, define

$$C_h(\hat{v}, a, w)_i = \begin{cases} (\tilde{T}_h^a w)_i, & |\hat{v}_i - (\tilde{T}_h^a w)_i| > \tau_h, \\ \hat{v}_i, & \text{otherwise.} \end{cases} \quad (\text{A.29})$$

The comparison in the implementation uses an outward upper bound for the difference. A selected entry is assigned the target directly; it is not reconstructed by adding two cancelling floating-point numbers. The indexed corrections, their masks, and the uncorrected neural outputs are saved. The deployed continuation

is therefore a neural representation with a recorded nodal correction, not a raw network.

PROPOSITION A.1—Evaluation guard: Suppose $\|\tilde{T}_h^a w - T_h^a w\|_\infty \leq \eta_h$. Then

$$\|C_h(\hat{v}, a, w) - T_h^a w\|_\infty \leq \tau_h + \eta_h.$$

For a frozen sequence of policies over N stages, with exact terminal data, the guarded backward values and the exact values of those policies differ by at most $\sum_{j=0}^{N-1} q_h^j(\tau_h + \eta_h)$. If $Nh = T$, $\tau_h = 0.1h^2$, and $\eta_h = o(h)$, this evaluation perturbation vanishes as $h \downarrow 0$. This statement is a property of the output rule, not a rate for the neural optimizer or an assertion of policy optimality.

The guard gives an implementable meaning to the evaluation component of a vanishing-perturbation condition. It does not supply the action-improvement error, the consistency error of a stopped diffusion approximation, or convergence of an uncorrected network away from the nodes. Its cost can be substantial: in the numerical economy it corrects a large fraction of fitted nodal values. We consequently count the correction memory and compare against an actor-free method with the identical guard. The construction is not evidence of neural compression when almost every node is retained.

A.1.7.2. Signed enclosures for continuous actions

Fix a state-time node and a continuation vector w . Write $Q_w(a)$ for its one-step payoff. The action set is the full compact box of the preference economy. Let $B = [\ell, r]$ be an action rectangle with centre c . A natural interval calculation supplies an upper bound $U^{\text{nat}}(B)$, while independent feasible evaluations supply a lower incumbent L .

The continuation is bilinear within state cells and continuous across their internal faces. Reflection changes its one-sided slopes. A stopped transition is different: the implementation switches between interpolated continuation and the specified stopping payoff, which need not agree away from boundary grid points. We call B regular when every cubature transition stays entirely inside, or entirely outside, each

wealth stopping face. Only on such a box do we use derivative information. All intersected interpolation cells and both sides of reflection folds enter the derivative hull.

Suppose $G_j(B) = [g_j^-, g_j^+]$ encloses every one-sided partial derivative with respect to action component j on a regular box. The mean-value upper enclosure is

$$U^{\text{mv}}(B) = \overline{Q}_w(c) + \sum_j \max\{g_j^-(\ell_j - c_j), g_j^-(r_j - c_j), g_j^+(\ell_j - c_j), g_j^+(r_j - c_j)\}, \quad (\text{A.30})$$

where $\overline{Q}_w(c)$ is an outward point upper bound. The algorithm uses the smaller of this bound and $U^{\text{nat}}(B)$. If $g_j^- > 0$, it restricts the supremum to $a_j = r_j$; if $g_j^+ < 0$, it restricts it to $a_j = \ell_j$. These are verified reductions to a maximizing face, not decisions based on a sampled derivative. For boxes crossing a stopping switch, the natural union enclosure is retained without this reduction.

THEOREM A.2—Signed action cover: *Assume that the natural enclosures contain their respective payoffs, that the derivative hulls contain all one-sided derivatives on each regular box, and that feasible point values and arithmetic bounds are outward. Start from a cover of the action box and apply subdivision, reduction to a verified maximizing face, and removal of a box only when its upper bound is at most $L + \varepsilon$. At any interruption, retain the upper bound of every unresolved box. Then*

$$\sup_{a \in \mathcal{A}} Q_w(a) \leq \max\{L + \varepsilon, \max_{B \text{ unresolved}} U(B)\}. \quad (\text{A.31})$$

The bound remains valid when a depth, work, or time budget is exhausted. If r action coordinates have verified strict derivative signs throughout a regular parent box, maximization over that box reduces to a face of dimension at most $d_a - r$.

The last statement describes a source of computational improvement, not a dimension-free worst-case bound. Under a K -Lipschitz gradient, a regular stationary box of diameter s has mean-value excess of order Ks^2 when its point evaluation error is negligible. Boxes that cross stopping switches, have ambiguous derivative signs, or contain many interpolation cells can remain expensive. Our experiment

measures their actual cost and keeps unresolved bounds; it does not assume that every box has favorable structure.

The operational NBO output is now explicit: the frozen policy representation, its guarded continuation vector when used, an independently propagated policy interval, and a complete-action optimal upper envelope. A nodal lookup policy is the object certified in the preference experiment. An off-grid interpolation of that policy is a different deployment rule and is evaluated separately. No online search is charged to a stored lookup table; constructing that table includes every actor, critic, and local-search query.

A.1.8. *Cost, Refinement, and Economic Accuracy*

The preceding applications define the original economic scope of NBO. We now evaluate the guarded map and signed cover on the same preference economy, and connect the high-dimensional experiment to an absolute economic benchmark. The comparison retains every earlier failure. In particular, a new upper envelope does not turn an actor-plus-search policy into an actor-only policy, and retraining on a finer grid does not repair the off-grid performance of a previously frozen policy.

A.1.8.1. *A tighter certificate for identical frozen policies*

The reference specification has 17×25 state nodes, twenty dates, and the continuous action box $[0.02, 0.30] \times [0, 1.5] \times [-0.45, 0.45]$. The ratios constrain consumption, leverage, and preference adjustment in the declared economy. They are economic primitives, not bounds on an unconstrained optimum. Active consumption or portfolio limits remain meaningful features of this constrained specification; changing them changes the economy.

We freeze the nine previously computed actor-plus-search, direct-plus-search, and search-only policies, including their original seeds. The complete-action upper envelope is recomputed independently of these policies' lower payoff envelopes. The new cover uses a one-step tolerance of 0.0002 and retains every unresolved upper bound. Its output can be shared across policies only because their action set, transition rule, domain, and payoff normalization coincide.

TABLE A.10

COMPLETE-ACTION BOUNDS FOR THE SAME FROZEN POLICIES

Policy	Seed	Uniform bound	Centre bound	≤ 0.1
Actor + search	11	0.0687	0.0068	Yes
Actor + search	29	0.0553	0.0050	Yes
Actor + search	47	0.0552	0.0075	Yes
Direct + search	11	0.0465	0.0071	Yes
Direct + search	29	0.0536	0.0052	Yes
Direct + search	47	0.0466	0.0075	Yes
Search only	11	0.0633	0.0113	Yes
Search only	29	0.0655	0.0053	Yes
Search only	47	0.0631	0.0076	Yes

Full 17×25 state grid, twenty dates, unchanged three-control box. The common signed cover took 372.53 seconds; 20 of twenty one-step covers closed at the requested tolerance. Other levels retain all unresolved upper bounds. Bounds are conditional on the stated outward-arithmetic execution model; continuous-state/time transfer is not included.

Table A.10 reports the resulting all-date, all-node loss bounds. The improvement relative to the historical table concerns the tightness of the verification calculation, not additional training or a change in preferences. The raw actors remain separately reported in the machine-readable common accounts. Search remains consequential, and these results do not establish an actor advantage over the actor-free comparator. They do show that the earlier failure of a 0.1 upper-bound target need not imply that the frozen hybrid policies themselves are more than 0.1 from the nodal optimum.

For a controlled cost comparison, Table A.11 gives one-step corner-only and signed covers at three dates, with identical continuation arrays, initial feasible actions, tolerance, depth limit, and work budget. Wall time includes derivative hull construction for the signed cover. The point of this experiment is not to compare two historical machines. Historical training costs are retained in the supplementary total-cost accounts with that qualification. Shared verification is amortized under explicitly stated populations of one, nine, and one hundred policies, and the welfare conversion has a separately counted cost. No claim of neural speed advantage is inferred from this accounting.

TABLE A.11

MATCHED ONE-STEP ACTION-COVER COMPUTATION

Cover	Date	Seconds	Q boxes, m.	Bracket	Budget hit
Corner	19	18.67	0.414	0.01336	Yes
Signed	19	9.50	0.066	0.00020	No
Corner	10	19.89	0.440	0.02230	Yes
Signed	10	14.60	0.112	0.00020	No
Corner	0	18.32	0.402	0.03430	Yes
Signed	0	25.84	0.446	0.00173	Yes

Continuation, incumbent, tolerance, and budget are identical within each date. Wall time includes gradient work; a query count is not a hardware-independent speed measure.

A.1.8.2. *The whole-domain refinement failure and its repair*

A frozen policy has to be distinguished from a policy refitted on a new grid. The complete historical refinement arrays reveal large losses away from the centre. Their maximum positive reference advantages, locations, quantiles, boundary-region summaries, and action saturation rates are retained in the complete numerical archive indexed in Appendix S.29. The largest historical discrepancy is approximately 2.1234 at high risk aversion and wealth close to the lower stopping boundary. Centre values alone do not summarize that experiment. These discrepancies are feasible-reference lower bounds on policy regret, not upper bounds for the continuous economy.

The new refinement comparison uses genuinely nested spatial grids, 17×25 and 33×49 , with twenty and forty dates, respectively. Each declared actor and actor-free search case is trained afresh. The network width, activation, actor budget, critic budget, and seed set are unchanged across the relevant ablation. We compare the unguarded fitted critic with the indexed guard in (A.29). For the forty-date case the target tolerance is 0.0000625. Both the fraction of corrected nodes and their storage cost are reported, rather than being attributed to an uncorrected neural approximation.

A new complete-action envelope is computed on the fine grid, not interpolated from the coarse certificate. It is applied to all twelve fine-grid policies, including

TABLE A.12

FRESH POLICIES ON THE NESTED FINE GRID

Method	Uniform bound range	Ref. advantage	Mean train s.	Passes
Actor	0.0654–0.0963	0.0932	88.32	3/3
Actor + guard	0.0140–0.0356	0.0155	87.75	3/3
Search	0.0655–0.0963	0.0932	51.07	3/3
Search + guard	0.0349–0.0357	0.0156	47.40	3/3

Full 33×49 state grid, forty dates; seeds 11, 29, and 47. Reference advantage is a lower regret diagnostic against the feasible 175-action reference, not an upper bound. Uniform bounds use a separately computed complete-action fine-grid envelope. The guard is a saved indexed correction, not an uncorrected network.

the six unguarded runs. This separates three facts: the failure of a frozen coarse-grid policy; the accuracy achieved by retraining within a declared nodal economy; and the additional error required to transfer a result to a continuously monitored diffusion. The first remains in the record, the second is measured, and the third is not assigned a zero value. Two levels alone do not establish an asymptotic convergence rate.

The deployment specification is a stored state-time action table with its saved neural proposal and continuation correction. Evaluating this table at its nodes requires no online maximizing search. Interpolating the table at off-grid states invokes a different policy rule; the historical experiment shows why this distinction matters. A raw actor without a table or local improvement has a different memory requirement and is not represented by the hybrid loss bounds.

A.1.8.3. *A model-specific welfare account*

An absolute utility gap is not a consumption-equivalent percentage when the utility exponent is itself a controlled state. We therefore compute a defined compensating transfer. Hold the saved policy and its state law fixed, leave preference-adjustment costs and the stopping bequest unchanged, and add externally financed consumption so that flow consumption becomes $(1 + e)c_t X_t$. This intervention changes felicity but does not charge its financing to wealth. It is a compensation experiment, not another admissible self-financed policy in the original box.

Let J_e^π be the compensated payoff and U the common optimal upper envelope. A verified lower envelope $J_e^\pi \geq U$ at every active starting node and date implies that this transfer compensates the policy's loss everywhere in the declared nodal economy. We search for a common e using monotonicity of CRRA felicity in consumption and verify the final inequality by outward propagation. Table A.13 gives the resulting upper percentages. These conservative transfers can be economically substantial even when the absolute utility bound is below 0.1; the threshold is not described as a universally small welfare loss.

TABLE A.13

SUFFICIENT FINANCED FLOW-CONSUMPTION SUPPLEMENTS

Proposal	Seed	Supplement, percent	Verification seconds
Actor	11	20.93	27.42
Actor	29	19.17	26.86
Actor	47	18.87	26.94
Direct	11	15.33	27.06
Direct	29	18.06	28.34
Direct	47	15.33	27.50
Search	11	20.44	31.32
Search	29	22.46	31.35
Search	47	19.42	31.37

Uniform sufficient compensation over every active starting node and date. State dynamics, the original policy, terminal bequest, and adjustment cost stay fixed. Consumption is externally financed. These are not feasible self-financed policy improvements or common homothetic consumption equivalents.

A.1.8.4. A global benchmark for the coupled capital economy

A comparison of two frozen policies does not indicate how far either is from the economic optimum. We supplement the paired simulations by a bound on the original, continuous-time, dense nonlinear capital economy. It does not rely on a Riccati equation, separability of the optimal policy, a state mesh, or a sampled residual.

In the notation of the capital experiment, let Y_i be log capital and $m_i \in [\underline{m}, \bar{m}]$ the consumption ratio. The flow payoff is $\overline{\log m} + \bar{Y} - (\zeta/2)\bar{m}^2$, and the terminal payoff is $\bar{Y} - \chi(\bar{Y} - \bar{Y})^2$, with $\chi = 0.03$. The drift of Y_i is $a + \kappa \tanh((BY)_i) - m_i - (\sigma_I^2 + \sigma_C^2)/2$. Common noise affects the mean but cancels from cross-sectional dispersion. Define

$$b_{\pm} = a \pm \kappa - \frac{1}{2}(\sigma_I^2 + \sigma_C^2), \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)},$$

and let

$$m_0(t) = \text{clip}_{[\underline{m}, \bar{m}]} \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}.$$

This common, deterministic schedule is feasible in the unchanged economy.

PROPOSITION A.3—Continuous-economy performance benchmark: *Starting from $Y_i(0) = 0$ for every i , let V^* be the optimal expected payoff of the original d -dimensional capital economy. Define*

$$U = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_+ - m_0(t)] \} dt,$$

$$D_d = \{ \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \}^2,$$

$$L_d = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_- - m_0(t)] \} dt - \chi e^{-\rho T} D_d.$$

Then $L_d \leq J(m_0) \leq V^* \leq U$ for every admissible dense coupling matrix B in the stated model. The inequalities apply to the continuous economy on $\mathbb{R}^d$, not merely to an Euler simulation.

The bound exploits bounded nonlinear drift, concavity of consumption utility, and an explicit feasible policy. Its numerical integration uses outward rectangle sums, interval exponential and logarithm evaluations, and square roots checked by interval squaring. Table A.14 reports the computed brackets. Their width is economically material and is not presented as a small regret bound for a learned

policy. They nevertheless provide an absolute benchmark absent from a paired-policy comparison. Simulations of m_0 use the same Brownian paths as the saved neural policies. The resulting differences and confidence intervals concern their common Euler discretization; an unknown simulation bias is not subtracted from the analytical bound to create a neural-policy certificate.

TABLE A.14

GLOBAL BOUNDS IN THE ORIGINAL CONTINUOUS CAPITAL ECONOMY

Dimension	Time panels	Feasible lower	Optimal upper	Width
10	4096	-1.489064	-1.051803	0.437261
10	16384	-1.488883	-1.051998	0.436884
20	4096	-1.489130	-1.051803	0.437327
20	16384	-1.488949	-1.051998	0.436950

Initial log capital is zero in every coordinate; horizon one. The bounds cover every admissible control on $\mathbb{R}^d$. They are not small loss bounds for neural policies. The same feasible schedule is evaluated on the saved Brownian paths in the supplement.

A.1.9. Benchmark Derivations

A.1.9.1. Merton consumption and transversality

Write $b = 1 - \gamma$ and $V = KX^b/b$. Then $V_X = KX^{-\gamma}$ and $V_{XX} = -\gamma KX^{-\gamma-1}$. Substituting $c = mX$ into the stationary time-additive equation and dividing by the positive number KX^b gives

$$0 = \frac{m^b}{bK} - \frac{\rho}{b} + r + p(\mu - r) - m - \frac{1}{2}\gamma p^2 \sigma^2.$$

Consumption maximization gives $m^{-\gamma} = K$; portfolio maximization gives $p = (\mu - r)/(\gamma\sigma^2)$ when interior. With a no-short-sale constraint and negative excess return, the maximizing share is zero. Let $R_* = r + p(\mu - r) - \gamma p^2 \sigma^2/2$. After substitution, $\gamma m = \rho + (\gamma - 1)R_*$.

For the optimal constant shares, $e^{-\rho t} \mathbb{E}[X_t^{1-\gamma}]$ decays as e^{-mt} . Thus $m > 0$ gives the candidate's additive transversality condition. Verification against a larger policy

class also requires the relevant localization and integrability conditions for the competitors, as specified in the main text. Policies with minus-infinite utility are not preferred to the finite candidate. The numerical laboratory checks the candidate formulas and does not silently apply a stationary constant to an unspecified finite-horizon terminal problem.

A.1.9.2. *Recursive coefficient and the utility interval*

For $f(c, V) = \rho[c^a(bV)^{1-a/b} - bV]/a$, the same homothetic substitution gives

$$0 = \frac{\rho}{a}[m^a K^{-a/b} - 1] + R_* - m.$$

Since $f_c = V_X$ implies $\rho m^{a-1} K^{-a/b} = 1$, the first term becomes $(m - \rho)/a$. Consequently, $m = \psi\rho + (1 - \psi)R_*$ and $K = [\rho m^{a-1}]^{b/a}$. The formula assumes the stationary candidate has positive consumption. It is not an infinite-horizon utility-selection theorem.

For the finite-horizon matching-bequest test at $a > 0 > b$, define $Y = bV$. The consumption domain is compact and bounded away from zero. At sufficiently small positive Y , $c^a Y^{-a/b} < 1$ uniformly in consumption, so $f < 0$. At sufficiently large Y , the reverse inequality holds and $f > 0$. Because dividing by negative b reverses the order of the utility endpoints, the large- Y constant is a lower utility barrier and the small- Y constant is an upper utility barrier. The matching bequest lies between them after enlarging the interval. Lipschitz scalar comparison then keeps each policy utility inside the interval, on which the aggregator and its utility derivative are bounded. This verifies the finite-horizon interpretation without claiming an unrestricted infinite-horizon result at those parameters.

A.1.10. *Endogenous-Preference Details*

A.1.10.1. *State restrictions and the reference experiment*

In the baseline application, the risk-aversion interval lies strictly above one and the process is reflected. The reference wealth problem is stopped at positive finite bounds. Consumption shares are bounded away from zero, adjustment efforts are bounded, and the horizon is finite. These choices bound the flow utility, its relevant

derivatives away from the terminal-boundary compatibility issue, and the bequest. They provide a well-defined finite reference calculation. The numerical reflection is a folded reflection of Brownian cubature endpoints; the wealth boundary is absorbing with the prescribed payoff. The same terminal and boundary values are imposed for both adjustment-cost parameters.

The grid reference operates in $y = \log X$. Its drift is

$$b_y = r + p(\mu - r) - m - \frac{1}{2}p^2\sigma_X^2,$$

and its covariance with u is $\rho\sigma_u p\sigma_X$. The two Brownian-factor columns can be written $(\sigma_u, \rho p\sigma_X)^\top$ and $(0, \sqrt{1 - \rho^2} p\sigma_X)^\top$. Four symmetric cubature directions reproduce this covariance. The correction $-p^2\sigma_X^2/2$ in log wealth is essential; omitting it would change the economic diffusion.

For each action, the backward update is current utility times the time step plus the exponentially discounted positive-weight interpolation of next-period value. Feasible actions are enumerated on the declared 27-element grid. Because the terminal/exit payoff is fixed, raising k weakly lowers each action's payoff and, by backward induction, lowers value at every state and time. The recorded arrays permit a direct check of this order. The reported grid-refinement difference is not an estimate of unrestricted neural approximation error.

A.1.10.2. Policy conditions and comparative statics

The consumption marginal utility in reference units is $U_c = \bar{c}^{-1}(c/\bar{c})^{-u}$. Equating it to V_X yields $c = \bar{c}(\bar{c}V_X)^{-1/u}$. Adjustment maximizes $\theta V_u - k\theta^2/2$ and hence uses the projection of V_u/k under a box. Portfolio choice maximizes a quadratic with coefficient $X^2\sigma_X^2 V_{XX}/2$. The projected first-order solution is valid only when this coefficient is negative. These conditions clarify why numerical curvature and feasibility must be checked independently.

The cross derivative V_{uX} is an endogenous object. Its sign cannot be inferred from the label “risk aversion” or from a negative shock correlation. Nor can the sign of $\partial_k \theta^k$ be inferred by differentiating $1/k$ while keeping V_u^k fixed. In contrast, the value monotonicity in k holds policy by policy. At a differentiable optimum the envelope formula follows because k appears only in the quadratic running cost. Changing

the normalization function $\kappa(u)$ can change both adjustment and portfolio policies even while leaving utility curvature in consumption unchanged.

An observation model is needed for structural estimation. With latent u , uncertainty in κ , and unknown adjustment or shock parameters, a sensitivity plot for one policy component does not prove identification. A practical estimation exercise would compute moment derivatives jointly, inspect their rank after normalization restrictions, and propagate numerical approximation error into the implied moment map. Those are distinct tasks from solving the controlled economy.

A.1.11. Temporal-Self Deviations

For the two-exponential kernel, the component value under constant consumption share m is $W_j(X) = m^b X^b / [b\lambda_j(m)]$, where $\lambda_j(m) = \rho_j - b(r - m)$. A deviation to share z for duration ϵ , followed by m , has component value at $X = 1$

$$J_j(z, \epsilon) = \frac{1}{b} \left[z^b \frac{1 - e^{-\lambda_j(z)\epsilon}}{\lambda_j(z)} + e^{-\lambda_j(z)\epsilon} \frac{m^b}{\lambda_j(m)} \right].$$

The continuous extension is used if $\lambda_j(z) = 0$. The deviation gain is $\sum_j w_j J_j(z, \epsilon) - \sum_j w_j W_j(1)$, with weights $(\beta, 1 - \beta)$. This is the quantity independently maximized in the experiment, rather than a restatement of the equilibrium first-order condition.

At an equilibrium share, the first-order gain is nonpositive for every feasible instantaneous share and equals zero at the equilibrium share. A finite-duration self can generally choose a slightly different share, so the maximized gain need not be exactly zero for positive duration. The required diagnostic is its gain divided by duration as duration tends to zero. For $\beta = 1$ the ordinary exponential solution is recovered.

As κ grows, the kernel becomes increasingly concentrated near the present before approaching its long-run component. Taking the instantaneous-deviation limit and taking $\kappa \rightarrow \infty$ are conceptually different operations. The finite- κ model in this paper explicitly specifies the order: the spike condition is derived for fixed κ , and any duration-limit comparative static is taken afterward. This avoids assigning an undefined continuous-time meaning to a discrete present-bias multiplier.

A.1.12. *Games and Independent Best Responses*

A fixed rival profile turns each player's problem into an ordinary controlled Markov problem, so the main verification results apply player by player. The derivative graph must implement this reduction. During player i 's actor step, every rival policy and every derivative of player i 's critic is detached. During player i 's critic step, all players' actions are detached. A sum of these reporting losses can be logged, but differentiation of a joint-profit objective is not equivalent to those block steps.

The two-firm static diagnostic has best response $q_i = (1 - q_j)/2$ when interior. At the joint-profit allocation $(1/4, 1/4)$, the deviating firm chooses $3/8$ and improves profit from $1/8$ to $9/64$, a gain of $1/64$. The own-action derivative is $1/4$ at the joint optimum. These numbers verify a graph-level distinction; they are not a computation of the dynamic capital game.

For the dynamic problem, a reported equilibrium should include an independent solution of each fixed-rival best-response problem, or an upper bound on the unilateral Hamiltonian gap over the entire verification domain. A finite list of unsuccessful deviations does not supply such an upper bound. Multiple initializations, asymmetric profiles, boundary actions, and the market-size normalization are part of the same experiment. The specification in the main text retains the dynamic Cournot application but does not invent a dynamic Nash table from the static arithmetic.

A.1.13. *Proof and Implementation of the Continuous-Action Envelope*

We use the notation of the main text's budget-safe envelope proposition. The nodal economy is fixed before verification. Its terminal and stopping payoffs, primitive coefficients, interpolation abscissae, and compact action box are part of the object being bounded. This is stronger than checking many action samples and weaker than verifying the continuous-state diffusion without a discretization account.

A.1.13.1. Complete-cover invariant and backward proof

At a given state and time, the initial action box covers the entire feasible set. Bisection replaces one closed box by two closed boxes with the same union. Their shared face is harmless. Induction on the number of subdivisions preserves coverage. A pruned box is retained logically through the inequality $u_B \leq c_i + \tau$, where u_B is an outward upper enclosure on that whole box, and c_i is a lower enclosure at a feasible incumbent. If the incumbent later increases, the previous pruning inequality remains true. Consequently every feasible action is either in an unresolved box or in a pruned box with payoff at most the current threshold. Taking the maximum of the threshold and all unresolved upper bounds dominates the action supremum, whether or not the supremum is attained. Termination on a numerical budget does not change this argument.

At N , the terminal inequalities are given. If $V_{n+1}^* \leq U_{n+1}$, positivity gives $T_n V_{n+1}^* \leq T_n U_{n+1} \leq U_n$. If $L_{n+1} \leq V_{n+1}^\pi$, the same argument gives $L_n \leq T_n^{\pi_n} L_{n+1} \leq T_n^{\pi_n} V_{n+1}^\pi = V_n^\pi$. Feasibility implies $V_n^\pi \leq V_n^*$. This proves the main envelope inequality. Substochasticity further gives

$$\|T_n w - T_n z\|_\infty \leq q_n \|w - z\|_\infty.$$

Applying this inequality to $U_n - T_n V_{n+1}^*$ yields the stated excess recursion and its product-sum expansion. The proof uses $q_n > 0$, not a stationary contraction assumption. Outward addition is used for the pruning threshold and final subtraction. The implementation records a requested tolerance and a retained optimization bracket separately. A budget-exhausted bracket is never replaced by that tolerance. *Q.E.D.*

A.1.13.2. Interpolation, reflection, stopping, and arithmetic

For a cell with coordinates $(u_0, u_1) \times (y_0, y_1)$, the continuation is

$$Iw(u, y) = (1 - s)(1 - t)w_{00} + s(1 - t)w_{10} + (1 - s)tw_{01} + stw_{11},$$

$$s = \frac{u - u_0}{u_1 - u_0}, \quad t = \frac{y - y_0}{y_1 - y_0}.$$

For fixed u , this is affine in y , so its extremes on a subrectangle occur at the two y endpoints. At either endpoint it is affine in u , so the extremes occur at the two u end-

points. Applying this argument to every intersected cell proves the corner enclosure. Nonnegative interpolation weights also permit interval nodal inputs: evaluating the corner expressions outward encloses every admissible nodal realization. A separate valid enclosure evaluates the rectangle center and adds cell-slope bounds over every intersected cell. Slopes are computed from stored nodal differences, not estimated from random derivative samples.

Preference reflection is piecewise affine. Its extrema on an interval occur at the interval endpoints or a crossed folding point. The allowed one-step movement is checked to exclude unhandled multiple folds. For log wealth, boxes fully inside the domain use interpolation, boxes fully outside use the prescribed clipped exit payoff, and boxes crossing an exit face use the union of both. On the stored grid the boundary nodes equal the exact prescribed payoff; at an off-grid exit the analytic payoff need not equal its nodal interpolant. The union construction avoids an unjustified global smoothness or Lipschitz assumption across that switch.

The implementation inherits the archived outward elementary and transcendental routines. Products, quotients, powers through logarithm and exponential, and endpoint rounding are enclosed. The logarithm's input is bounded below by 0.02; the preference denominator is separated from zero because $u \in [1.2, 3]$. The recorded control model treats the stored binary64 primitive coefficients and abscissae as exact real inputs. The evidence also tests agreement with the inherited training evaluator on the finite action grid. Such a regression check is useful for implementation consistency but is not the proof of global action coverage. That proof is the maintained union invariant and the interval inclusion property. The arithmetic account is conditional on the specified floating-point execution model, not a formal verification of hardware, libraries, or the Python interpreter.

A.1.13.3. *A common upper envelope for multiple policies*

The optimal upper array depends on the economy, not on which feasible incumbent produced the lower search values. Thus the completed upper array can be reused for any frozen feasible policy on the same state, time, and continuous-action domains. We separately propagate each policy's lower and upper payoff arrays and subtract its lower array from the common optimal upper array. Costs are reported

both as shared verification and as the cost of an individual policy evaluation. Sharing a verifier does not make its cost vanish. It also does not justify transferring the upper array to a changed wealth box, action constraint, utility normalization, time step, or interpolation rule.

A.1.14. Separating Numerical and Continuous-Economy Error

The following result specifies the transfer work needed in addition to a nodal certificate. It is not used to set an uncomputed transfer term to zero. Let $\mathcal{T}_n$ be the exact economic one-step optimality operator with its continuously monitored stopping and reflection rule, and let $\mathcal{T}_n^\pi$ be the operator for a declared continuous-state implementation of the numerical policy. Let I_n be a positive, sup-norm-nonexpansive interpolation from grid values to the economic state domain. Suppose the exact operators are q_n -Lipschitz on the comparison class.

PROPOSITION A.4—Transfer with separate optimal and policy defects: *Suppose terminal interpolation error is at most B and, along the relevant nodal continuations,*

$$\|\mathcal{T}_n I_{n+1} v_{n+1}^* - I_n T_n v_{n+1}^*\|_\infty \leq d_n^*,$$

$$\|\mathcal{T}_n^\pi I_{n+1} v_{n+1}^\pi - I_n T_n^\pi v_{n+1}^\pi\|_\infty \leq d_n^\pi.$$

Define $D_N^* = D_N^\pi = B$ and $D_n^j = d_n^j + q_n D_{n+1}^j$ for $j \in \{*, \pi\}$. Then the exact economic values satisfy

$$0 \leq \mathcal{V}_n^* - \mathcal{V}_n^\pi \leq I_n(U_n - L_n) + D_n^* + D_n^\pi.$$

The action-domain approximation belongs in d_n^* , while policy implementation, boundary monitoring, state interpolation, and time/cubature errors may affect both defects.

PROOF: Insert $\mathcal{T}_n I_{n+1} v_{n+1}^*$ between the exact and interpolated nodal Bellman values. The Lipschitz property bounds the propagated value difference by $q_n D_{n+1}^*$, and the displayed defect bounds the remaining difference by d_n^* . Backward induction gives $\|\mathcal{V}_n^* - I_n v_n^*\| \leq D_n^*$. The same argument gives the fixed-policy bound. Subtract the two decompositions and use positivity of I_n and $v_n^* - v_n^\pi \leq U_n - L_n$. Feasibility gives the lower inequality. Q.E.D.

The proposition can use local regularity, coupling, barrier, or weak-approximation estimates to bound its defects; it does not claim that those hypotheses hold automatically for a fitted critic near a stopped boundary. In the present two-state computations, action coverage is operationally bounded on the nodal economy. The continuous-state/time defects remain explicitly uncomputed. Nested grids and common frozen policies diagnose sensitivity but do not provide the one-sided inequalities required by this proposition.

TABLE A.15: Frozen continuous-policy refinement, common economic domain

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	11	17×25	20	-4.843442	-4.841008
Actor	11	25×37	40	-4.848086	-4.839462
Actor	11	33×49	80	-4.842578	-4.830165
Actor	29	17×25	20	-4.841652	-4.841008
Actor	29	25×37	40	-4.846754	-4.839462
Actor	29	33×49	80	-4.840763	-4.830165
Actor	47	17×25	20	-4.844135	-4.841008
Actor	47	25×37	40	-4.848828	-4.839462
Actor	47	33×49	80	-4.843387	-4.830165
Direct	11	17×25	20	-4.843773	-4.841008
Direct	11	25×37	40	-4.857445	-4.839462
Direct	11	33×49	80	-4.852920	-4.830165
Direct	29	17×25	20	-4.841889	-4.841008
Direct	29	25×37	40	-4.854488	-4.839462
Direct	29	33×49	80	-4.848755	-4.830165
Direct	47	17×25	20	-4.844185	-4.841008
Direct	47	25×37	40	-4.854178	-4.839462
Direct	47	33×49	80	-4.848680	-4.830165

Notes. Policies are frozen on the original time schedule and bilinearly interpolated in state. References still use 175 actions. Differences are sensitivity diagnostics, not continuous-state/time upper bounds.

A.1.15. *Strategic Deviations and Missing Pure Stages*

For a fixed Markov rival profile in a finite-horizon finite-state/action game, condition on any history. The own player's conditional payoff from a randomized current action is a convex combination of pure-action payoffs with the same continuation best-response values. It cannot exceed the largest such payoff. Backward induction therefore bounds every randomized and history-dependent unilateral policy by the Markov pure-action best-response recursion. Outward propagation then provides a bound over this full deviation class, not merely the particular local deviations observed during actor training.

Existence of a pure equilibrium in a fitted one-step game is a different matter. If none exists, minimum-defect selection produces a feasible profile, not an exact equilibrium. The independent dynamic best response still evaluates that profile correctly. The following records give every missing pure stage in the six principal finite-policy study neural runs, including time index, state, and minimum maximum stage defect. Indices begin at zero. The six runs have $6 \times 30 \times 225 = 40,500$ active state-time nodes; the five recorded exceptions are not excluded from their final dynamic bounds.

The profit targets are absolute units of the displayed per-firm payoff. Raw network policy, shortlist policy, and corrected finite lookup policy are different deployment objects. The principal finite-policy study guard fraction is zero, but the full joint-action oracle was nevertheless constructed during training. A mixed-strategy *profile* was not estimated; the preceding argument explains why mixed unilateral deviations against the fixed pure profile are already covered by the finite best-response bound.

TABLE A.16

EVERY MISSING PURE FITTED ONE-STEP EQUILIBRIUM IN FINITE-POLICY STUDY

Market	Seed	Time index	K_1	K_2	Minimum defect
1	11	1	0.140625	0.865625	6.492e-06
1	11	1	0.684375	0.140625	4.018e-07
1	11	10	0.321875	0.140625	1.538e-05
2	11	16	0.231250	1.046875	1.534e-05
2	29	13	0.140625	0.956250	1.394e-06

Minimum defect is the smallest maximum unilateral one-step improvement over all joint finite actions. These nodes remain in full dynamic verification; they are not called exact stage equilibria.

A.1.16. Paired Payoffs, Costs, and Economic Scaling

Each row below compares the two frozen methods on the same stored Brownian paths, at the indicated dimension, seed, fitting domain, and Euler resolution. Intervals are pointwise Student intervals across 512 pathwise differences. They are not independent across step sizes, because the Brownian increments are nested; no simultaneous-coverage claim is made. The narrow and wide policies were separately fitted, so their comparison also changes the training domain and update budget.

TABLE A.17: All method-paired policy-payoff comparisons

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
10	11	Narrow	40	-0.03043	-0.03234	-0.02852
10	11	Narrow	80	-0.03237	-0.03438	-0.03036
10	11	Narrow	160	-0.03338	-0.03544	-0.03133
10	11	Wide	40	0.00061	0.00058	0.00064
10	11	Wide	80	0.00055	0.00052	0.00057
10	11	Wide	160	0.00052	0.00049	0.00054
10	29	Narrow	40	-0.02800	-0.02946	-0.02655
10	29	Narrow	80	-0.02993	-0.03146	-0.02840
10	29	Narrow	160	-0.03094	-0.03250	-0.02938

	Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper	
1								1
2								2
3	10	29	Wide	40	-0.00007	-0.00008	-0.00006	3
4	10	29	Wide	80	-0.00004	-0.00005	-0.00003	4
5	10	29	Wide	160	-0.00003	-0.00004	-0.00002	5
6	10	47	Narrow	40	-0.02522	-0.02695	-0.02349	6
7	10	47	Narrow	80	-0.02675	-0.02856	-0.02493	7
8	10	47	Narrow	160	-0.02755	-0.02941	-0.02569	8
9	10	47	Wide	40	0.00004	0.00003	0.00005	9
10	10	47	Wide	80	0.00001	0.00001	0.00002	10
11	10	47	Wide	160	0.00000	-0.00001	0.00001	11
12	20	11	Narrow	40	-0.05798	-0.06054	-0.05543	12
13	20	11	Narrow	80	-0.06131	-0.06396	-0.05866	13
14	20	11	Narrow	160	-0.06302	-0.06572	-0.06033	14
15	20	11	Wide	40	0.00123	0.00118	0.00128	15
16	20	11	Wide	80	0.00097	0.00092	0.00101	16
17	20	11	Wide	160	0.00084	0.00080	0.00088	17
18	20	29	Narrow	40	-0.01308	-0.01346	-0.01269	18
19	20	29	Narrow	80	-0.01401	-0.01442	-0.01361	19
20	20	29	Narrow	160	-0.01449	-0.01490	-0.01408	20
21	20	29	Wide	40	0.00480	0.00463	0.00496	21
22	20	29	Wide	80	0.00432	0.00417	0.00448	22
23	20	29	Wide	160	0.00409	0.00394	0.00424	23
24	20	47	Narrow	40	-0.19715	-0.19765	-0.19666	24
25	20	47	Narrow	80	-0.20011	-0.20061	-0.19962	25
26	20	47	Narrow	160	-0.20161	-0.20211	-0.20111	26
27	20	47	Wide	40	0.00084	0.00078	0.00089	27
28	20	47	Wide	80	0.00066	0.00061	0.00071	28
29	20	47	Wide	160	0.00058	0.00053	0.00062	29

Notes. Pointwise 95 percent Monte Carlo intervals, 512 paired paths per row. They do not bound optimality, continuous-time bias, or simultaneous coverage.

The additional representation comparisons below use the identical NDU finite model. Gated regression has more parameters and higher per-update work than

the width-32 tanh critic. Tensor projection exploits low state dimension. Both facts are explicit rather than hidden by matching only nominal iteration counts.

TABLE A.18

ADDITIONAL CONTINUATION REPRESENTATIONS, IDENTICAL FINITE PREFERENCE ECONOMY

Method	Seed	Parameters	Policy bound	Value error	Train (s)	Verify (s)
Chebyshev	11	121	0.0121	0.0775	0.14	4.41
Chebyshev	11	225	0.0039	0.0165	0.15	4.48
Chebyshev	11	49	0.0373	0.2325	0.14	4.36
Gated	11	9345	0.0079	0.0284	41.28	4.12
Gated	29	9345	0.0098	0.0200	42.59	4.49
Gated	47	9345	0.0088	0.0306	38.99	4.30

Gated denotes a DGM-style architecture fitted to finite Bellman targets, not a differential DGM solver. Chebyshev degrees are 6, 10, and 14. All actions are enumerated; construction and reference costs are separately retained in JSON. Deterministic projections are not replicated under fictitious random seeds.

The finite-policy study NDU actor’s correction map stores a small number of changed action indices, but its complete policy table occupies 68,000 bytes with the archived int64 representation. A minimal pair of int64 state-time and replacement indices requires 16 bytes per correction, excluding indexing and container overhead. A stored-table deployment avoids actor evaluation at existing nodes; a raw-network deployment requires its network weights and argmax calculation. Neither observation establishes off-grid policy or guard correctness. The continuous hybrid additionally requires local action queries using the stored continuation representation.

In the continuous-action study, training-query counts exclude verification queries by construction. All-action references are created only after policies and critics have been frozen. The raw actor, direct-gradient, multi-head, and smoke failures are retained with their full configurations. The complete action verifier’s tighter and looser implementations have separate records, including unresolved boxes and declared budgets. A passing regression suite means that arithmetic, scope, and

accounting checks passed; it does not change a failed economic accuracy target into a success.

TABLE A.19
RETAINED CONTINUOUS-ACTION DEVELOPMENT FAILURES

Configuration	Seed	Deviation gain	Train (s)	Target 0.1
Actor smoke	11	0.1872	0.53	Fail
Raw actor	11	0.2596	15.77	Fail
Four-head reset	11	1.1033	16.83	Fail
Direct gradient	11	0.1997	16.07	Fail

All executed pilot sources, raw arrays, and histories remain. The smoke grid and budget differ from the principal grid; it is not a principal-seed replicate. Pilot selection and subsequent improvements are explicitly post-review development.

A.1.16.1. Complete configuration records

The principal application comparisons above retain the whole-domain failure, fresh-grid comparison, policy correction costs, and consumption compensation. The complete configuration tables additionally report every boundary-region summary, action-saturation rate, training seed, correction record, and capital payoff pair. They remain in the exact numerical archive identified in Appendix S.29; the editorial map records the original table labels and source lines. A policy refitted on a nested grid remains a different object from a frozen policy evaluated off grid. None of those negative results is replaced by the successful comparison at the center of a state domain.

REFERENCES

HERDEGEN, MARTIN, DAVID HOBSON, AND JOSEPH JEROME (2021): “The Infinite Horizon Investment-Consumption Problem for Epstein–Zin Stochastic Differential Utility,” ArXiv:2107.06593v1. [5]
SIRIGNANO, JUSTIN AND KONSTANTINOS SPILIOPOULOS (2018): “DGM: A Deep Learning Algorithm for Solving Partial Differential Equations,” *Journal of Computational Physics*, 375, 1339–1364. [13]